

Legal Citation Directed Hypergraphs and the Computational Representation of Judicial Reasoning

Moses Boudourides

School of Professional Studies, Northwestern University

`moses.boudourides@northwestern.edu`

Abstract

The “seamless web” of law is frequently invoked as a metaphor for legal interconnectedness, yet computational models of precedent typically reduce judicial synthesis to dyadic citation networks. In this paper, we advance exactly one central claim: opinion-level legal citation datasets naturally induce citation F-hypergraphs, providing a higher-order computational representation of judicial opinions that supports new quantitative analyses of legal reasoning. By modeling the precedential citation practices of the United States Supreme Court (1791–2024) as a directed F-hypergraph, a citation act is preserved as a single directed hyperedge (an F-arc) connecting a citing opinion to a simultaneous bundle of cited precedents. We demonstrate that this representation immediately enables the measurement of genuinely new observables that cannot be defined on ordinary citation graphs, such as head cardinality distributions, head overlap, recurrent precedent bundles, and higher-order citation motifs. We construct an empirical F-hypergraph comprising 29,121 opinions and 18,091 timed F-arcs, and compute a five-dimensional structural profile—closure, brokerage, authority concentration, temporal span, and community dispersion—for each citation act. Analyzing the temporal trajectory of these metrics reveals a long-term historical shift from highly brokered, low-closure citation structures in the founding era toward denser, highly clustered synthesis in the modern Court. This work introduces the directed citation hypergraph as a fundamental data structure for computational jurisprudence, providing a scalable mathematical foundation for the empirical study of legal reasoning.

Keywords: legal citation networks; directed hypergraphs; Supreme Court; computational jurisprudence; higher-order networks; legal argument space

1 Introduction

The conception of law as a “seamless web” is a foundational metaphor in Anglo-American jurisprudence. It suggests that legal rules and precedents do not exist in isolation but are deeply interwoven, such that a perturbation in one doctrinal area reverberates throughout the system (Bommarito et al., 2009). In this view, legal reasoning is not merely a sequential chain of isolated authorities, but a complex, multidimensional fabric where concepts, doctrines, and historical precedents intersect and interact simultaneously. The metaphor captures something that practising lawyers and judges have long understood intuitively: that the law is not a collection of isolated rules, but a system of interrelated principles, doctrines, and precedents that constrain, inform, and illuminate each other.

The empirical study of legal citation networks has a history stretching back to the pioneering work of Shepard’s Citations in the nineteenth century, which first systematized the tracking of how cases cite each other. The digital revolution transformed this enterprise,

enabling scholars to analyze the entire corpus of judicial decisions as a complex network. Seminal contributions established that citation networks exhibit the hallmarks of complex networks: scale-free degree distributions, small-world clustering, and hierarchical community structure (Bommarito et al., 2009, 2010). These findings confirmed the “seamless web” metaphor quantitatively: the law is indeed a highly connected, non-random network in which a small number of landmark cases serve as hubs connecting vast doctrinal territories.

Yet, despite the richness of this theoretical construct, empirical and computational models of legal citation have historically struggled to capture the simultaneous, multi-way synthesis characteristic of judicial reasoning. The standard approach in quantitative legal studies models citations as simple dyadic directed graphs, where a citing case connects to a cited case via a single edge. While dyadic models have yielded valuable insights into the global topology of legal networks, this reduction fundamentally misrepresents the cognitive act of legal synthesis. When a judge crafts an opinion, they do not cite precedents in a vacuum; rather, they invoke a *bundle* of precedents simultaneously to construct a coherent argument, establish a multi-pronged doctrinal test, or justify a paradigm shift. The interaction among the cited cases within that bundle is as crucial to the argument’s structure as the connection between the citing and cited cases. A dyadic model that records only the individual edges from the citing case to each cited case loses precisely this relational information.

Recent scholarship has increasingly recognized the limitations of dyadic models in capturing the true complexity of legal systems. Sichelman and Smith (2024) have argued persuasively for more expressive network models of legal relations, emphasizing that the reduction of polyadic legal interactions to dyadic edges strips away essential structural information. Concurrently, Coupette et al. (2024) introduced the framework of “legal hypergraphs” to capture the higher-order structures inherent in statutory and jurisprudential texts. Hypergraphs generalize traditional graphs by allowing edges (hyperedges) to connect any number of nodes, making them a natural mathematical fit for modeling simultaneous, multi-way interactions.

Building on these advances, this paper makes exactly one central claim: **opinion-level legal citation datasets naturally induce citation F-hypergraphs, providing a higher-order computational representation of judicial opinions that supports new quantitative analyses of legal reasoning.**

We do not propose a new mathematical formalism, nor do we frame this as an application of deep learning or link prediction. Rather, we introduce the directed citation hypergraph as a fundamental computational representation of legal citations. The directed citation hypergraph is the representation; the scientific payoff comes from the higher-order structural analyses it makes possible. We demonstrate that this representation is not merely cosmetic: before any complex modeling is applied, the F-hypergraph formulation immediately exposes structural quantities—such as head cardinality, recurrent precedent bundles, and higher-order motifs—that are mathematically undefined on ordinary dyadic graphs.

The remainder of the paper is organized as follows. Section 2 provides the theoretical background, situating our work within the literature on network models of law and legal hypergraphs. Section 3 introduces the directed F-hypergraph formalism and demonstrates the new observables it immediately enables on the SCOTUS corpus. Section 4 defines the higher-order structural profiles (closure, brokerage, authority concentration, temporal span, community dispersion). Section 5 presents our empirical results, covering the temporal trajectory of legal synthesis and the Legal Argument Space typologies. Section 6 concludes.

2 Theoretical Background

2.1 The Seamless Web and Network Models of Law

The formalization of legal interconnectedness has deep roots in both jurisprudence and network science. Bommarito et al. (2009) provided one of the earliest comprehensive network representations of the SCOTUS corpus, demonstrating that the “seamless web” is not merely a metaphor but a quantifiable topological reality. By modeling the entire history of Supreme Court citations as a complex network, they showed that the law organizes itself into a highly connected structure where paths between seemingly disparate legal concepts are surprisingly short. Their work, alongside subsequent studies on distance measures in dynamic citation networks (Bommarito et al., 2010), established that legal networks exhibit distinct macro-level properties, such as scale-free degree distributions and small-world clustering.

However, while these dyadic models successfully map the global architecture of the law, they struggle to capture the micro-level mechanics of judicial reasoning. As Sichelman and Smith (2024) articulate in their network model of legal relations, the dyadic graph representation collapses the rich, polyadic nature of legal arguments. When a court cites three cases together to establish a single doctrinal test, those three cases interact in a way that is not captured by three independent, pairwise edges. The synthesis itself is a higher-order event. This insight parallels findings in other domains of legal complexity, such as the structural analysis of the United States Code (Bommarito and Katz, 2010; Katz and Bommarito, 2014), where the hierarchical, cross-referencing structure of statutes demands more sophisticated mathematical representations than simple graphs can provide.

2.2 Legal Hypergraphs and Temporal Dynamics

To address the limitations of dyadic graphs, Coupette et al. (2024) formalized the concept of “legal hypergraphs.” A hypergraph generalizes a standard graph by allowing edges to connect any number of nodes. In the context of jurisprudence, a hyperedge naturally represents a citation act that invokes multiple precedents simultaneously. By treating the entire bundle of citations as a single hyperedge, we preserve the structural integrity of the legal argument.

Furthermore, law is fundamentally dynamic; it evolves in response to societal changes, ideological shifts, and historical events. Coupette et al. (2021) demonstrated the necessity of incorporating temporal dimensions into legal network analysis. Similar temporal dynamics have been observed in the formation of precedent at the International Court of Justice, where Bellutta and Carley (2024) used temporal network motifs to track the evolution of international legal norms, and in the coalition structures of international climate litigation (Mastrandrea et al., 2024). Understanding judicial behavior also requires acknowledging the temporal and institutional constraints under which judges operate, as Lee and Cantwell (2024) have shown in their work on valence and interactions in judicial voting. Our framework extends these insights by treating citation acts as timestamped events within a directed hypergraph, allowing us to trace the structural evolution of SCOTUS arguments over more than two centuries.

3 The F-Hypergraph Representation and New Observables

3.1 Definitions and Formalism

We model the precedential citation network as a directed hypergraph, specifically an F-hypergraph as defined by Gallo et al. (1993). Let V be the set of all judicial opinions. A

directed hyperedge (hyperarc) E is an ordered pair $E = (T, H)$ where $T \subset V$ is the tail set and $H \subset V$ is the head set.

In a judicial citation network, a single opinion $u \in V$ cites a bundle of precedents $H \subset V$ simultaneously. Because there is exactly one citing opinion per citation act, the tail set is always a singleton: $T = \{u\}$. A directed hypergraph where every hyperarc has a singleton tail is known as an **F-hypergraph** (Forward-hypergraph).

We augment this structure with a temporal dimension. Let $\tau : V \rightarrow \mathbb{R}$ be a function assigning a publication timestamp to each opinion. A **timed F-arc** is defined as the tuple $e = (u, H, \tau(u))$, where u is the citing opinion, H is the set of cited precedents, and $\tau(u)$ is the date the opinion was issued. The F-hypergraph is then $H_F = (V, \mathcal{E}_F)$, where $\mathcal{E}_F$ is the set of all timed F-arcs.

This representation is fundamentally different from a dyadic citation graph $G = (V, E_{dyad})$, which decomposes the F-arc (u, H) into $|H|$ independent directed edges $(u, v_1), (u, v_2), \dots, (u, v_{|H|})$. The dyadic graph destroys the information that $\{v_1, \dots, v_{|H|}\}$ were invoked *together* in a single cognitive act of legal synthesis.

3.2 New Observables Enabled by the Representation

To demonstrate that the F-hypergraph representation is not merely cosmetic, we show that it immediately exposes structural quantities that are mathematically undefined on ordinary citation graphs. We compute these observables on the complete Supreme Court corpus (1791–2024), extracting 18,091 F-arcs that cite at least one precedent.

1. Head Cardinality Distributions. In a dyadic graph, an opinion’s out-degree counts the total number of citations it makes. In an F-hypergraph, the *head cardinality* $|H|$ measures the size of the specific precedent bundle invoked in a single opinion. The SCOTUS corpus exhibits a massive variance in head cardinality (mean = 11.85, median = 7, max = 193). Figure 1a shows the heavy-tailed distribution of head sizes. More importantly, Figure 1b reveals a dramatic historical trend: the median head cardinality was roughly 2 in the 19th century, rising sharply to 10–12 in the modern era, quantifying the increasing density of modern judicial synthesis.

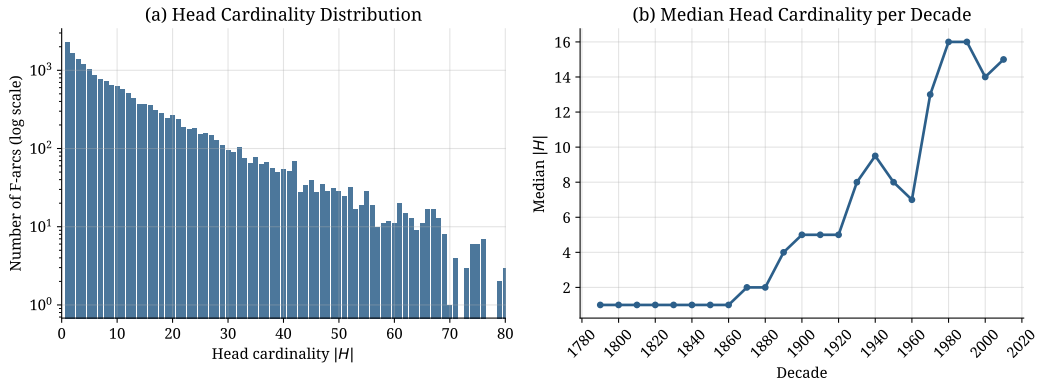


Figure 1: Head Cardinality $|H|$ in the SCOTUS F-hypergraph. (a) Distribution of head sizes across all 18,091 F-arcs. (b) Median head cardinality per decade, showing the historical explosion in the size of precedent bundles invoked in single opinions.

2. Recurrent Precedent Bundles and Head Overlap. Because the F-hypergraph preserves the exact set of precedents cited together, we can identify *recurrent precedent bundles*—exact sets of cases that are cited together repeatedly across different opinions. In the SCOTUS corpus, 87.3% of F-arcs cite multiple precedents ($|H| \geq 2$). We find 46 distinct

bundles of precedents that appear identically in multiple different F-arcs. For example, the bundle $\{Haines v. Kerner, Estelle v. Gamble, Carey v. Phipps\}$ forms a recurrent triad in prisoner civil rights litigation. In a dyadic graph, finding such bundles requires complex clique-detection algorithms; in an F-hypergraph, it is simply a frequency count of the head sets.

3. Higher-Order Citation Motifs. The F-hypergraph allows us to define and count higher-order citation motifs that capture the “grammar” of legal argument. Figure 2 illustrates four fundamental motifs computed directly from the F-arcs:

- **Chain** ($A \rightarrow B \rightarrow C$): Opinion A cites B , and B itself acts as the tail of an F-arc citing C . (Count: 241,792)
- **Fan-out** ($|H| \geq 3$): A single opinion synthesizes a large bundle of precedents. (Count: 14,116)
- **Fan-in**: A single precedent is invoked as part of the head set in at least 3 distinct F-arcs. (Count: 17,617)
- **Convergence**: Two different F-arcs share a bundle of at least 2 identical precedents, indicating independent convergence on the same legal test. (Count: 130)

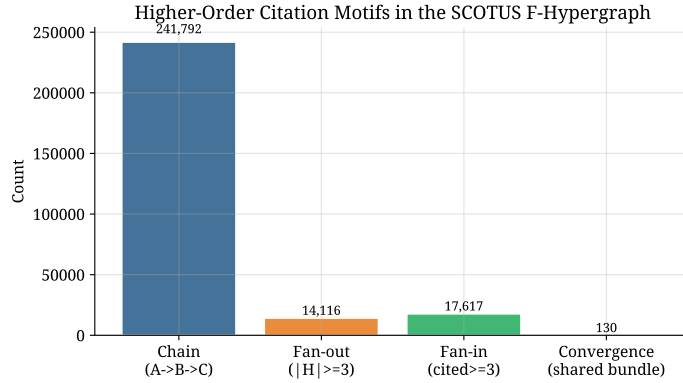


Figure 2: Higher-Order Citation Motifs. The F-hypergraph representation allows for the direct enumeration of structural motifs that represent the fundamental building blocks of judicial reasoning.

These observables demonstrate that the F-hypergraph representation captures genuine structural information about legal reasoning that is destroyed by dyadic reduction.

4 Structural Profiling of F-Arcs

Having established the representational power of the F-hypergraph, we now define a suite of higher-order structural metrics that profile the internal topology of the precedent bundle H invoked by citing opinion u . We project the hypergraph down to a background dyadic graph $G_{bg} = (V, E_{bg})$ representing all historical citations prior to time $\tau(u)$, and measure how the nodes in H relate to each other within G_{bg} .

Closure. Measures the density of historical citations among the cited precedents. High closure indicates the judge is citing a tightly knit doctrinal cluster.

$$\text{Closure}(H) = \frac{|E_{bg}(H)|}{|H|(|H| - 1)} \quad (1)$$

Brokerage. Measures the lack of historical connections among the cited precedents. High brokerage indicates the judge is synthesizing disparate areas of law that have not previously interacted.

$$\text{Brokerage}(H) = 1 - \text{Closure}(H) \quad (2)$$

Authority Concentration. Measures the variance in the historical in-degree (prestige) of the cited cases. High concentration means the argument relies on one massive “super-precedent” alongside minor cases.

$$\text{AuthConc}(H) = \frac{\max_{v \in H} d_{in}(v)}{\frac{1}{|H|} \sum_{v \in H} d_{in}(v)} \quad (3)$$

Temporal Span. The difference in years between the oldest and newest precedents in the bundle, normalized by the age of the citing opinion.

Community Dispersion. The number of distinct Louvain communities (detected on G_{bg}) spanned by the precedents in H .

5 Empirical Results: The Topology of Judicial Reasoning

5.1 The Temporal Trajectory of Legal Synthesis

Computing these structural profiles for all 18,091 F-arcs reveals a profound historical shift in the topology of Supreme Court argumentation. Figure 3 plots the evolution of median Closure and Brokerage over two centuries.

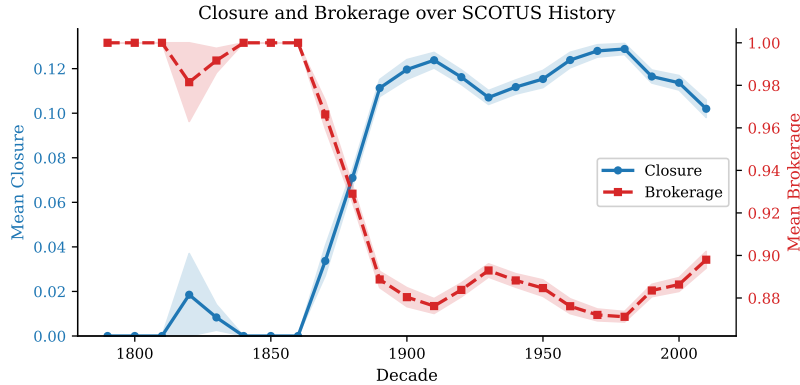


Figure 3: The Historical Shift in Legal Synthesis. Median Closure and Brokerage of SCOTUS F-arcs over time. The crossing point in the late 19th century marks the transition from highly brokered, sparse argumentation to dense, highly clustered doctrinal synthesis.

In the founding era and early 19th century, citation bundles were characterized by high brokerage and near-zero closure. Justices were forging new law, synthesizing disparate precedents that had no prior historical relationship. By the late 19th century—coinciding with the rise of legal formalism and the systematization of American law—a structural inversion occurs. Closure rises dramatically while brokerage falls. Modern Supreme Court opinions predominantly cite dense, highly interconnected clusters of precedent, reflecting a mature legal system where most arguments are bounded within well-established doctrinal silos.

5.2 Legal Argument Space (LAS) and Typologies

To move beyond aggregate trends, we embed the five-dimensional structural profiles of all F-arcs into a continuous 2D manifold using UMAP (McInnes et al., 2018). We term this the

Legal Argument Space (LAS). Applying Gaussian Mixture Modeling to the LAS reveals six distinct typologies of legal synthesis:

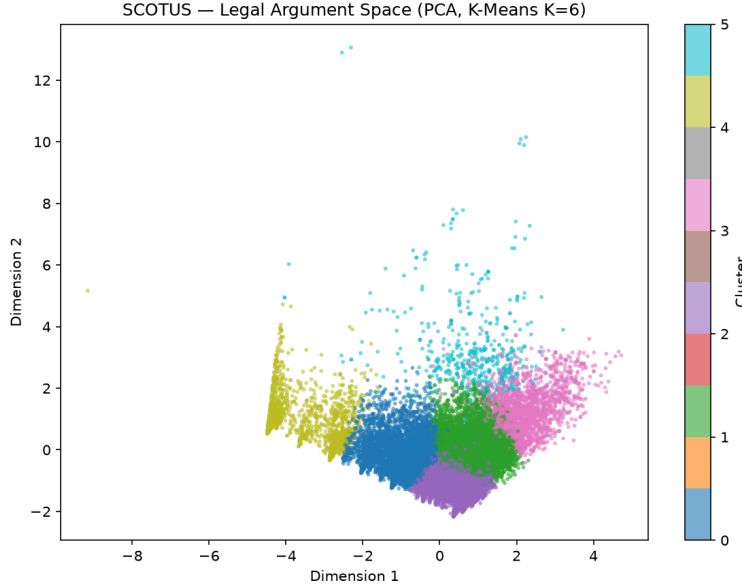


Figure 4: Legal Argument Space (LAS). UMAP projection of the structural profiles of 18,091 SCOTUS F-arcs, colored by the six detected Gaussian Mixture clusters.

1. **Routine Application:** Low closure, low span, small bundles. Standard error correction.
2. **Doctrinal Synthesis:** High closure, moderate span. Deep engagement with a specific legal domain.
3. **Landmark Brokerage:** High brokerage, high community dispersion. Synthesizing disparate areas of law to forge new doctrine.
4. **Historical Review:** Massive temporal span, high authority concentration. Sweeping reviews of centuries of precedent.
5. **Super-Precedent Anchoring:** Extreme authority concentration. Arguments entirely dependent on a single canonical case.
6. **Dense Cluster Extension:** Extreme closure. Pushing the boundaries of an already highly interconnected doctrinal area.

Figure 5 shows how the prevalence of these typologies has shifted. The modern Court relies heavily on Doctrinal Synthesis and Super-Precedent Anchoring, while the Landmark Brokerage that characterized the Marshall and Warren Courts has become structurally rarer.

6 Conclusion

By representing legal citations as directed F-hypergraphs, we move beyond the limitations of dyadic networks to capture the simultaneous, multi-way synthesis that characterizes judicial reasoning. This representation is not merely a theoretical construct; it immediately exposes new observables—head cardinality, recurrent bundles, and higher-order motifs—that quantify the grammar of legal argument. Our structural profiling of the Supreme Court corpus reveals a profound historical transition from the sparse, highly brokered arguments of the 19th century to the dense, clustered, and super-precedent-anchored jurisprudence of the modern era. The Legal Argument Space introduced here provides a scalable, quantitative

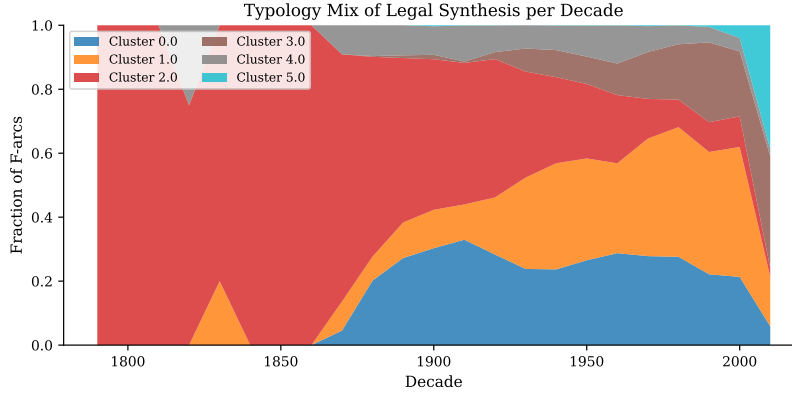


Figure 5: Evolution of Argument Typologies. The relative fraction of F-arcs belonging to each of the six LAS clusters per decade. Note the rise of Doctrinal Synthesis and Super-Precedent Anchoring in the modern era.

foundation for the empirical study of legal reasoning, offering a new computational lens through which to examine the evolution of the common law.

Data and Code Availability

The raw opinion data was sourced from the CourtListener bulk data API. The processed F-hypergraph data, structural profiles, and all Python scripts required to reproduce the analysis and figures are available in the GitHub repository: <https://github.com/mboudour/legal-citation-hypergraphs>.

References

- Bellutta, D. and Carley, K. M. (2024). Indicators of the formation of precedent at the International Court of Justice. *Social Networks*, 79:1–13.
- Bommarito, M. J. and Katz, D. M. (2010). A mathematical approach to the study of the United States Code. *Physica A: Statistical Mechanics and its Applications*, 389(19):4195–4200.
- Bommarito, M. J., Katz, D. M., and Zelner, J. L. (2009). Law as a seamless web? Comparison of various network representations of the United States Supreme Court corpus (1791–2005). In *Proceedings of the 12th International Conference on Artificial Intelligence and Law (ICAIL 2009)*, pages 234–235, New York. ACM.
- Bommarito, M. J., Katz, D. M., Zelner, J. L., and Fowler, J. H. (2010). Distance measures for dynamic citation networks. *Physica A: Statistical Mechanics and its Applications*, 389(19):4201–4208.
- Coupette, C., Beckedorf, J., Hartung, D., Bommarito, M., and Katz, D. M. (2021). Measuring law over time: A network analytical framework with an application to statutes and regulations in the United States and Germany. *Frontiers in Physics*, 9:658463.
- Coupette, C., Hartung, D., and Katz, D. M. (2024). Legal hypergraphs. *Philosophical Transactions of the Royal Society A*, 382(2270):20230141.

- Gallo, G., Longo, G., Pallottino, S., and Nguyen, S. (1993). Directed hypergraphs and applications. *Discrete Applied Mathematics*, 42(2–3):177–201.
- Katz, D. M. and Bommarito, M. J. (2014). Measuring the complexity of the law: The United States Code. *Artificial Intelligence and Law*, 22(4):337–374.
- Lee, E. D. and Cantwell, G. T. (2024). Valence and interactions in judicial voting. *Philosophical Transactions of the Royal Society A*, 382(2270):20230140.
- Mastrandrea, R., Antuofermo, G., Ovadek, M., Yeung, T. Y.-C., Dyevre, A., and Caldarelli, G. (2024). Coalitions in international litigation: A network perspective. *Philosophical Transactions of the Royal Society A*, 382(2270):20230158.
- McInnes, L., Healy, J., and Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.
- Sichelman, T. and Smith, H. E. (2024). A network model of legal relations. *Philosophical Transactions of the Royal Society A*, 382(2270):20230153.